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AI and Machine Learning Research

Perception, Prediction, Nowcasting and Decision Support for Low-Visibility Haulage

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ABSTRACT

This document sets out where machine learning genuinely contributes to the proposed system and, equally important, where it does not. It opens by drawing a firm line between learned components and the deterministic safety core, then covers radar perception networks, multimodal fusion under sensor degradation, trajectory prediction on a constrained road network, fog nowcasting, alert-logic tuning, operational optimisation and predictive maintenance. It closes with the data strategy, model lifecycle, edge deployment constraints and validation approach, together with an explicit statement of the failure modes that machine learning introduces into a safety system.

Prepared as part of a solution-design research package addressing dense-fog and low-visibility haul road operations in large-scale mechanised open cast iron ore mines. Figures relating to NMDC production, statutory provisions and commercial systems are compiled from public sources; site-specific values marked as indicative require validation against NMDC operational records before use in a formal proposal.

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1. Where ML Belongs and Where It Does Not

Machine learning is central to several capabilities in this system and dangerous in others. The distinction should be made architecturally, at the outset, rather than being negotiated case by case during development.

Function	Approach	Rationale
Range, velocity, angle measurement	Deterministic signal processing	Physics-based, analysable, verifiable. No learning required or desirable.
Position computation and integrity bound	Deterministic estimation (Kalman family)	Requires provable error bounds. A learned position estimate cannot supply a protection level.
Detection of a return above the noise floor	Deterministic (CFAR)	Statistically characterisable false alarm rate.
Object classification	Learned	Class boundaries are not analytically specifiable.
Clutter and ghost rejection	Hybrid	Deterministic geometric rules, refined by learning.
Trajectory prediction	Hybrid	Physics-based kinematics constrained by the road model, refined by learned behavioural patterns.
Time-to-collision computation	Deterministic	Simple kinematics; must be auditable after an incident.
Final alert decision	Deterministic rules on fused evidence	Must be explainable, reproducible and testable. This is the safety core.
Alert threshold tuning	Learned offline, deployed as fixed parameters	Learn the tuning, deploy it deterministically.
Fog nowcasting	Learned	Complex local meteorology; no tractable analytical model.
Fleet sequencing and dispatch optimisation	Learned / optimisation	Not safety-critical; benefits from learning.
Predictive maintenance	Learned	Not safety-critical.

The governing principle: a learned component may add information to the safety chain, but it must never be the sole basis for withholding a warning. Classification improves alert quality and reduces nuisance alerts; a low-confidence or unclassified detection must still be treated as a hazard. If the classifier says "probably a rock" and the kinematics say "closing at 8 m/s in lane", the kinematics decide. This asymmetry - ML can escalate freely, but can de-escalate only within bounds - keeps the system's failure behaviour conservative and its decisions auditable.

2. Radar Perception

2.1 Detection and classification on sparse radar point clouds

4D imaging radar produces a sparse, noisy point cloud - typically hundreds to a few thousand points per frame, against the hundreds of thousands a LiDAR produces. The sparsity and the presence of Doppler as a per-point attribute make it a distinct problem from LiDAR perception, and architectures transfer only with adaptation.

Applicable architectures.

- ? **Point-based networks** (PointNet/PointNet++ lineage) operating directly on unordered point sets, extended to take Doppler as an input channel alongside spatial coordinates and intensity. Well suited to sparse input.
- ? **Pillar and voxel encoders** (PointPillars lineage) projecting points into a bird's-eye-view grid and applying 2D convolution. Efficient on embedded hardware; bird's-eye view is also the natural frame for a haul road, where the problem is essentially planar.
- ? **Range-Doppler map networks** operating on the radar data cube before detection, applying 2D convolution to the range-Doppler representation. Retains information that thresholding discards, at higher compute and data-rate cost.
- ? **Transformer-based detectors** for sparse point sets, offering strong performance at higher compute cost. A Phase 3 consideration rather than an initial deployment choice.

Recommendation. A pillar-based bird's-eye-view detector with Doppler as an input channel, augmented by temporal accumulation across several frames to densify the input. Temporal accumulation is particularly valuable for radar: a target that is sparse in one frame becomes well defined across five, and ego-motion compensation using the precise L1 position solution makes accumulation accurate rather than smeared. **This is a concrete instance of the positioning layer improving the perception layer.**

2.2 Micro-Doppler pedestrian classification

The highest-consequence classification task is separating a person from a similarly sized rigid object. Radar offers a strong physical discriminator: a walking person's limbs produce a time-varying Doppler signature - micro-Doppler - distinct from the single Doppler component of a rigid body.

Approach. Compute a spectrogram of the Doppler return over a short window at each candidate detection, and classify the resulting time-frequency pattern with a compact convolutional network. This is well established in the radar literature and is computationally modest.

Value here. It provides pedestrian classification using a modality that works at MOR 3-5 m, where the camera and thermal classifiers that would normally perform this task are degraded or blind. Given that the run-over of a person by a reversing or manoeuvring truck is among the most severe accident archetypes in open cast mining, and that fog makes it more likely, this is a high-priority capability.

2.3 Clutter and ghost rejection

Volume III Document 1, Section 6 identifies ground clutter and multipath ghosting as the principal radar challenges in a ferrous pit. A hybrid approach is recommended:

- ? **Deterministic first pass.** Reject returns geometrically inconsistent with the road model; reject zero-relative-Doppler returns consistent with the known ego-velocity; apply the dynamically estimated ground plane from IMU pitch and modelled gradient.
- ? **Learned second pass.** A classifier trained on site-collected data distinguishing genuine targets from residual clutter and ghosts, using features including kinematic plausibility, track persistence, return statistics, spatial relationship to known geometry, and consistency with the cooperative layer.

Data requirement. This capability cannot be developed without Bailadila data. Multipath behaviour against high-grade haematite faces is site-specific. This is a direct argument for the Phase 0 instrumented vehicle campaign (Volume III Document 1, Section 12, item V3).

3. Multimodal Fusion Under Degradation

The key requirement is that fusion must adapt to sensor validity rather than assume it. The published literature provides the pattern directly: architectures that integrate LiDAR with long-range 77 GHz and short-range 24 GHz radar plus stereo and thermal cameras, with an explicit **performance assessment layer** that monitors

LiDAR degradation and shifts reliance to complementary sensors when performance falls below a critical level. Related work on FMCW LiDAR and 77 GHz radar fusion employs an **adaptive fusion kernel** balancing LiDAR spatial precision against radar penetration as visibility decreases, reporting sustained detection beyond 100 m where LiDAR alone fails beyond approximately 70 m.

Recommended approach for this system.

- ? **Explicit sensor validity estimation.** Each sensor reports its own current reliability: radar reports radome attenuation and interference level; camera and thermal report contrast, blur and aperture obscuration; GNSS reports protection level. These are measured, not inferred from the weather.
- ? **Visibility-conditioned weighting.** Fusion weights are a function of measured per-zone MOR (from the L0 visibility network) *and* of each sensor's self-reported validity. Two independent signals govern the weighting, so a fouled radar in clear weather is handled as correctly as a clean radar in dense fog.
- ? **Track-level fusion,** chosen for robustness to individual sensor failure, ease of validation, and the ability to add or remove sensors across a phased, heterogeneous rollout.
- ? **Graceful degradation with explicit declaration.** When fusion quality falls, the system declares it - satisfying FR-17 and HR-11, and directly addressing failure modes F2, F3 and F5.

The essential discipline: the system must know what it cannot currently see, and must say so. A fusion architecture that silently produces a confident-looking output from degraded inputs is the most dangerous possible design, because it induces precisely the over-trust that turns an assistance system into a hazard.

4. Trajectory Prediction and Threat Assessment

Physics first. Constant-velocity and constant-turn-rate models, propagated against the **road model** rather than in free space, provide the baseline prediction. On a haul road this is unusually powerful: vehicles are strongly constrained to follow a known corridor, so a prediction that follows the modelled road is far more accurate than a straight-line extrapolation - particularly on the switchbacks where straight-line prediction is worst and the hazard is greatest.

Learned refinement. A behavioural model trained on the fleet's own trajectory history captures what physics does not: typical speed profiles by road segment, load state and gradient; braking behaviour on approach to known features; queueing patterns at loading and tipping points; the distinction between a truck slowing to turn and one slowing to stop.

Threat assessment. Deterministic computation on the predicted trajectories:

- ? Time-to-collision against each track, computed along the road geometry rather than in straight-line space.
- ? Time-to-lane-departure from the lateral position and rate.
- ? Time-to-boundary for geofenced zones, tipping edges and exclusion areas.
- ? Required deceleration to avoid, compared with achievable deceleration given load, gradient and surface state - the concept borrowed from ETCS braking curve supervision.

The alert decision is deterministic and auditable. Learned components supply inputs - a class, a confidence, a refined trajectory - but the decision to alert is a rule over those inputs. After an incident, it must be possible to state exactly why the system did or did not warn. A learned end-to-end alert decision cannot support that, which is why it is excluded.

5. Fog Nowcasting

Requirement FR-04 asks for a 15-30 minute visibility forecast per zone. This is one of the strongest applications of machine learning in the system, because local fog behaviour is complex, site-specific and not well served by general numerical weather prediction at this horizon.

Framing. Predict MOR, or the probability of falling below each class threshold, for each zone, at horizons of 15, 30 and 60 minutes.

Inputs. Current and recent MOR from the L0 network; temperature and dew point, with the spread being the primary physical predictor of condensation; relative humidity; wind speed and direction, which govern advection and dispersal; rainfall rate and recent history; barometric pressure; solar radiation, which governs radiation fog burn-off; time of day and day of year; and the spatial pattern across the sensor network, which carries information about an approaching fog bank that no single sensor holds.

Model options.

Approach	Assessment
Gradient-boosted trees on engineered features	Recommended starting point. Strong on tabular data, works with limited history, interpretable, cheap to train and deploy.
LSTM / GRU sequence models	Good for temporal dependency; needs more data.
Temporal convolutional networks	Efficient; good long-range temporal context.
Spatio-temporal graph networks	Models the sensor network as a graph, capturing advection between zones. The right target architecture once sufficient data exists.
Physics-informed hybrid	Combines a condensation physics model with learned residual correction. Attractive for generalisation from limited data.

The data constraint is decisive. Useful skill requires at least one, preferably several, monsoon seasons of observations. **This is the single strongest argument for deploying the visibility network immediately,** ahead of any other element of the programme (Volume III Document 2, Section 5). Every month of delay is a month of training data that cannot be recovered later.

Verification. Nowcast skill must be measured against outcomes using proper scoring - Brier score for probabilistic class forecasts, critical success index for threshold crossings - and reported as KPI K12. An unverified forecast that is trusted is worse than no forecast.

Operational value. The value is not in predicting fog for its own sake but in acting earlier. Knowing that a zone will drop to V3 in twenty minutes allows the control room to sequence trucks clear of the affected section, complete cycles in progress, and stage the transition - rather than having the fog arrive and stop the fleet where it stands. This addresses levers 3 and 4 of the loss model directly, and those levers, as Volume I Document 1 argued, are the most underestimated.

6. Alert Logic and False Alarm Reduction

Risk R3 - nuisance alerts causing operator disengagement - is rated the highest combined likelihood and impact in the programme risk register. Machine learning contributes to its mitigation, carefully.

The mechanism of nuisance alerts. A simple distance envelope alerts whenever anything is near, regardless of whether it is a threat. On a haul road, trucks are routinely near each other - queueing at a tip, passing on a two-way section, spotting at a face. A distance-based system alerts constantly in exactly the situations that occur most often, which is the fastest possible route to operator disengagement.

Layered mitigation.

- ? **Trajectory-aware alerting (FR-19).** Alert on time-to-conflict, not proximity. A truck 20 m ahead travelling at the same speed is not a threat; one 60 m ahead closing at 8 m/s is. This single change removes the large majority of nuisance alerts.
- ? **Context suppression.** Suppress alerts for interactions that are normal in context - a queue at a tipping point, a passing manoeuvre on a designated passing bay, a truck being loaded by a shovel - identified from the road model and the operating state broadcast in the V2V message.
- ? **Learned contextual modelling.** Learn, from logged data, which alerts operators consistently mark as nuisance, and refine suppression rules accordingly. Learn the normal interaction patterns at each location so that the abnormal stands out.
- ? **Operator feedback loop (HR-10).** Single-action nuisance reporting feeding the training data. This closes the loop and gives operators agency in tuning the system, which materially supports acceptance.
- ? **Graded alerting (FR-23).** Awareness, caution and warning levels so that most information is conveyed without an alarm.

The critical safety constraint on this work. Suppression logic reduces false alerts and can, if wrong, suppress a true one. Therefore:

- ? Suppression must be **deterministic and auditable**, learned offline and deployed as explicit rules, never as an opaque runtime model.
- ? Every suppression must be **logged** so that suppressed-but-real events are discoverable in review.
- ? Suppression must be **bounded**: no suppression below a floor time-to-collision, regardless of context. Below that floor, the system always warns.
- ? Suppression rule changes must be **validated against the historical event record** before deployment.

7. Operational Optimisation and Decision Support

Non-safety applications, where learning operates with fewer constraints:

- ? **Fleet sequencing under degraded visibility.** Given a visibility forecast, current fleet state and production targets, optimise dispatch to maximise throughput within the operating-class constraints. Reinforcement learning or constrained optimisation over a simulation of the haul network.
- ? **Restart optimisation (lever 4).** On release from a stoppage, sequence the fleet to restore throughput fastest - which is not the same as releasing everything at once, as that produces congestion at the loading faces and tipping points.
- ? **Speed advisory optimisation.** Recommend a speed profile per segment given measured visibility, validated sensing range, gradient, load and traffic, balancing cycle time against the stopping-distance constraint.
- ? **Production impact estimation (OR-05).** Estimate the production consequence of alternative fog responses so that the control room decision is quantified. This converts the suspension decision from reflexive to informed - a change of kind, not of degree.
- ? **Digital twin (OR-08).** A simulation of the haul network, fleet and weather supporting scenario analysis, operator training in simulated fog, and offline testing of dispatch and alert policies before they touch a real vehicle. The training application is particularly valuable: operators can experience the system in V4 conditions safely and repeatedly, building calibrated trust rather than discovering the system's limits on a live ramp.

8. Predictive Maintenance and System Health

Secondary but worthwhile:

- ? **Anomaly detection on system health telemetry**, identifying degrading units before they fail - addressing failure mode F3, silent failure.
- ? **Calibration drift detection** from the statistics of residuals between sensors and the road model, satisfying RR-06 and addressing F10.
- ? **Fouling rate prediction** from weather and operating history, scheduling cleaning before performance degrades rather than after.
- ? **Vehicle maintenance inference** from the rich motion data the positioning system collects - braking performance degradation being directly relevant to the stopping-distance assumptions on which alert thresholds are built. A truck whose measured deceleration has degraded should have its alert thresholds adjusted, which is a genuinely novel safety capability that falls out of the architecture.

9. Image Enhancement - A Cautionary Assessment

Dehazing, contrast enhancement, defogging networks and generative reconstruction are widely published and frequently proposed for exactly this problem. The assessment here is negative, and it is important to be explicit about why, because these methods are superficially compelling and demonstrate well.

Why they fail at MOR 3-5 m. Dehazing algorithms invert a physical model - typically an atmospheric scattering model - to recover scene radiance from an attenuated, veiled image. This works while the scene signal remains above the sensor noise floor. At an extinction coefficient of 600-1000 per km, the direct transmission term at even 20 m is attenuated by many orders of magnitude. **There is no signal to recover.** The algorithm has nothing to invert.

Why generative reconstruction is worse than useless here. A generative model asked to produce a plausible clear image from an opaque input will produce one - a hallucination consistent with its training distribution and with the few pixels of actual information available. Presented to an operator on a haul road, this is an actively dangerous artefact: a confident, detailed, plausible image of a road that may not match reality. **This must be excluded from the design by explicit prohibition**, not merely by omission, because it will be proposed.

Where enhancement is legitimate. In classes V0-V2, moderate haze enhancement improves the camera layer's contribution to classification, which supports the nuisance alert reduction of Section 6. It should be applied to the *machine* perception pipeline and must never be presented to the operator as a view of the world.

The correct response to "can we use AI to see through the fog?" is that the information is not present in the optical signal, so no algorithm can recover it - and that the architecture's answer is not to see through fog but to render the road from the map and detect the residual with radar. Any proposal to synthesise an operator-facing visual scene from an opaque image should be rejected on safety grounds.

10. Data Strategy

The programme's data assets, in priority order:

Dataset	Content	Why it matters	When to start
D1 Visibility and meteorology	Continuous MOR and met parameters per zone	Trains the nowcast; provides the acceptance reference; quantifies the problem	Immediately - Phase 0

Dataset	Content	Why it matters	When to start
D2 Radar in fog	Raw and processed radar, co-located with measured MOR, from an instrumented vehicle	Validates the central architectural assumption; trains clutter and ghost rejection	Immediately - Phase 0, monsoon-gated
D3 Production baseline	Fog stoppage hours, stop-start counts, fog-condition cycle times, from existing FMS and shift records	The business case and the KPI baseline	Immediately - Phase 0, from existing records
D4 Fleet trajectories	Precise position histories	Trains behavioural prediction; detects road model divergence	Phase 2
D5 Alert and event log	Every alert with full context, plus operator nuisance reports	Tunes alert logic; the core of R3 mitigation	Phase 1
D6 Multimodal perception	Synchronised radar, thermal, camera, cooperative, with annotation	Trains and validates detection and classification	Phase 1-2
D7 System health	Telemetry, faults, fouling, calibration residuals	Predictive maintenance; drift detection	Phase 1

Public datasets. Seeing Through Fog, RADIATE, Oxford Radar RobotCar and the multimodal vehicle detection derivatives enable algorithm development and architecture selection before any site data exists, which is a real de-risking opportunity. **They are adequate for development and inadequate for validation**, because none contains a loaded haul truck on a wet ore-fines ramp inside monsoon orographic cloud. Site data collection is a programme deliverable, not a convenience.

Annotation. Annotating radar data is harder than annotating images, because a human cannot easily interpret a radar point cloud directly. The practical approach is **cross-modal annotation**: collect radar together with LiDAR and camera in clear conditions, annotate the easily interpretable modalities, and transfer the labels to the radar. This yields a labelled radar dataset in clear conditions; the model then transfers to fog because **radar's own returns change little between clear and foggy conditions** - which is the whole premise of the architecture, and a fortunate property for data collection.

11. Edge Deployment Constraints

All safety-critical inference runs on the vehicle (principle P4, requirement CR-06), on automotive-grade hardware within thermal and power budgets far tighter than a development workstation.

Constraint	Implication
Latency budget of 200 ms end-to-end (PR-12)	Perception inference budget of roughly 50-60 ms; rules out large models
Automotive-grade SoC with NPU	Quantised models (INT8), operator sets restricted to what the NPU supports
Thermal limits in a hot cab	Sustained inference load must not thermally throttle; design to sustained, not peak, capability
Deterministic timing	Fixed inference cost per frame; no data-dependent runtime variation
Memory	Model size and activation memory bounded
No cloud fallback	Full capability offline, always

Practical approach. Compact architectures designed for edge deployment; quantisation-aware training; pruning; operator fusion; and a hard rule that the deterministic safety chain runs on a fixed, predictable budget independent of how much the learned components are doing in a given frame. If the perception network

overruns, the safety chain still executes on the evidence it has.

12. Model Lifecycle and Governance

- ? **Versioning.** Every deployed model is versioned, and every logged event records the model version that produced it. Without this, post-incident analysis is impossible.
- ? **Staged deployment.** Shadow mode first - the new model runs alongside the deployed one, its outputs logged but not acted upon, and the two compared on real data. Then limited fleet deployment, then full.
- ? **Regression testing.** A held-out corpus of safety-critical scenarios that every candidate model must pass. The corpus grows as real events occur: every near-miss and every incident becomes a permanent test case.
- ? **Drift monitoring.** Monitor input distributions and output statistics for drift from training conditions. A model trained on one season's fog may not suit another; a model trained before a haul road was re-cut may not suit after.
- ? **Change control.** Model updates that affect the safety chain are change-controlled and, where they alter safety behaviour, notified to DGMS under GR-05.
- ? **Rollback.** Every deployment must be reversible, quickly, across the fleet.

13. Validation and Assurance

Validating learned components in a safety system requires more than a test-set accuracy figure.

- ? **Performance against measured MOR (AR-02, AR-03).** Detection and classification performance reported across visibility classes, not as a single aggregate. An aggregate figure dominated by clear-weather data says nothing about the case that matters.
- ? **Stratified evaluation.** By target class, range band, visibility class, road geometry, time of day and load state. Aggregate metrics conceal exactly the failures that matter - a model can score well overall while failing on pedestrians at 25 m in V3.
- ? **Failure analysis over accuracy.** Characterise *how* the model fails and under what conditions, not merely how often.
- ? **Adversarial and edge-case testing.** Unusual targets, partially occluded targets, multiple simultaneous targets, targets at the boundaries of detection envelopes.
- ? **Explainability for the safety chain.** The deterministic alert decision must be reconstructible from logs. Learned inputs must be logged with their confidences so the decision can be reconstructed exactly.
- ? **Independent validation.** Evaluation by a party other than the developer, on data the developer has not seen.

14. Failure Modes Introduced by ML

Stating these explicitly is part of the assurance argument. Machine learning does not merely fail to work; it fails in characteristic ways that a deterministic system does not.

Failure mode	Mechanism	Mitigation
Distribution shift	Model performs well on training conditions and degrades on conditions not represented - a new fog regime, a re-cut road, a new machine class	Drift monitoring; stratified evaluation; site-specific training data; conservative fallback on low confidence

Failure mode	Mechanism	Mitigation
Confident error	Model outputs high confidence on a wrong classification	Never allow classification alone to de-escalate an alert; kinematics override class
Silent degradation	Performance falls gradually without a discrete failure	Continuous monitoring of output statistics; shadow-mode comparison against a reference
Rare-class failure	Under-represented classes - an unusual vehicle, a person in an unusual posture - are poorly detected	Deliberate over-sampling; targeted data collection; kinematic fallback for unclassified detections
Spurious correlation	Model learns a feature correlated with the label in training but not causal - e.g. associating a target class with a location where it was always observed	Careful dataset construction; stratified evaluation; failure analysis
Hallucination in generative components	A generative model produces plausible content unsupported by the input	Excluded by prohibition - no generative component in the operator-facing chain (Section 9)
Feedback loop contamination	The system changes operator behaviour, which changes the data, which re-trains the model on its own effects	Maintain a held-out reference corpus; monitor for behavioural shift; periodic re-baselining
Over-trust induced by fluency	A smooth, confident interface implies a capability the model does not have	Persistent display of effective sensing range and system health (FR-27, HR-11)

The last row is the most important and the least technical. A well-presented machine learning system communicates competence through the quality of its interface, independently of its actual performance. In a safety system used by operators who cannot verify its output against what they see - because they cannot see - that is a serious hazard. The architecture's insistence on persistently displaying what the system can and cannot currently see is the principal defence, and it should be treated as a safety requirement rather than an interface preference.